

Name	Aravinthaa S
Register No	230701032
Dept & Sec	CSE A

Ex No – 8	Producer Consumer Using Semaphores
-----------	------------------------------------

Program :

```
#include<stdio.h>

#include<stdlib.h>

int mutex = 1, full = 0, empty = 3, x = 0;

int wait(int s) {
    return (--s);
}

int signal(int s) {
    return (++s);
}

void producer() {
    mutex = wait(mutex);
    full = signal(full);
    empty = wait(empty);
    x++;
    printf("\nProducer produces the item %d", x);
    mutex = signal(mutex);
}

void consumer() {
    mutex = wait(mutex);
    full = wait(full);
    empty = signal(empty);
    printf("\nConsumer consumes item %d", x);
```

```

x--;

mutex = signal(mutex);
}

int main() {

    int n;

    printf("\n1. Producer\n2. Consumer\n3. Exit");

    while (1) {

        printf("\nEnter your choice: ");

        scanf("%d", &n);

        switch (n) {

            case 1:

                if ((mutex == 1) && (empty != 0))

                    producer();

                else

                    printf("Buffer is full!!!");

                break;

            case 2:

                if ((mutex == 1) && (full != 0))

                    consumer();

                else

                    printf("Buffer is empty!!!");

                break;

            case 3:

                exit(0);

                break;

            default:

                printf("Invalid choice!");

        }
    }
}

```

```
}  
  
return 0;  
  
}
```

Output :

```
1. Producer  
2. Consumer  
3. Exit  
Enter your choice: 1  
  
Producer produces the item 1  
Enter your choice: 2  
  
Consumer consumes item 1  
Enter your choice: 2  
Buffer is empty!!  
Enter your choice: 1  
  
Producer produces the item 1  
Enter your choice: 1  
  
Producer produces the item 2  
Enter your choice: 1  
  
Producer produces the item 3  
Enter your choice: 1  
Buffer is full!!  
Enter your choice: 3  
[cse32@localhost ~]$
```